GGG-V-410B
December 10, 1993
SUPERSEDING
GGG-V-410A
May 26, 1966

FEDERAL SPECIFICATION

VISE, BENCH, CLAMP BASE; SHEET METAL-WORKER'S; BENCH AND PIPE; JEWELER'S; MACHINIST'S; AND MULTIPOSITION

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers vises for use in metalworking trades.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. Metalworking vises covered by this specification shall be of the following types and classes as specified (see 6.2).

Type I	-	Bench, clamp base (utility)
Class 1	-	Stationary base
Class 2	-	Swivel base
Type III	-	Sheet metalworker's
Type IV	-	Bench and pipe (swivel base, stationary jaw)
Type V	-	Jeweler's
Class 1	-	Stationary base
Class 2	-	Swivel base
Type VI	-	Machinist's
Class 2	-	Swivel base, stationary jaw
Class 3	-	Swivel base, swivel jaw
Type VII	-	Multiposition bench

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2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

Federal Specification:

PPP-P-40	-	Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-H28	-	Screw-Threads Standards for Federal Services
FED-STD-123	-	Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM):

ASTM E 8	-	Methods of Tension Testing of Metallic Materials
ASTM E 18	-	Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of vises otherwise conforming to this specification.

3.2 Materials. The materials used in the construction of vises shall be as hereinafter specified for each type and class. All castings shall be of uniform quality, free of blowholes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects. Strength and other essential physical properties of the casting shall be adequate throughout for the purpose intended. When materials are used which are not speci-fically designated, they shall be entirely suitable for the purpose intended.

3.3 Coatings. All unmachined surfaces of the component parts of vises shall be given either one coat of enamel or lacquer. The color may be at the option of the contractor. In no case shall fillers be used.

3.4 Interchangeability of parts. All parts having the same manufacturer's number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.5 Components. The vise shall consist of a body with back jaw, front jaw, vise screw, nut, sliding member, as applicable, screw handle, and bench-clamping screw or base, as applicable, and jaw facings, when required.

3.5.1 Body. Except as otherwise specified herein, the body shall be of cast iron, ductile iron, malleable iron, steel casting, or steel forgings. The back jaw shall be an integral part of the body. The face of the back jaw shall be furnished square and perpendicular to the base, or the axis of the hole in the body through which the sliding member passes, as applicable, within a tolerance of 2 in either direction (see 4.5). The bottom surface of the body base shall be solid in cross section. When guide rods are furnished, the body shall be threaded to receive the vise screw. The length of the threaded bearing portion of the body shall be not less than one and one-half times the diameter of the screw.

3.5.2 Front jaw. The front jaw, except for jaw facings where specified, shall be of cast iron, ductile iron, malleable iron, steel casting, or steel forgings. When front jaws and sliding members are

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furnished, the sliding member shall conform to the requirements of 3.5.2.1. The front jaw shall be integral or attached to the sliding member. The face of the front jaw shall remain parallel to the face of the back jaw in any position of travel.

3.5.2.1 Sliding member. The sliding member may be either rectangular or round. It shall have a smooth machined or cold rolled finish to allow not more than 0.008 inch clearance at any point between the sliding member and its counterpart for type I, classes 1 and 2, and type V, class 1. Not more than 0.012 inch clearance shall be permitted for type IV; type V, class 2; and type VI, classes 2 and 3. The respective measurement shall be construed to mean a peripheral clearance of the sliding member and its counterpart with the clearance being measured simultaneously at four opposing points 90° to each other. The member shall slide in the body with smooth action without binding at any point throughout the full range of travel. Machined portions of sliding members shall be sufficiently rugged to withstand abrasion of filing, grindings, or impact from knocks to which they are normally subjected. Backlash in either direction shall not exceed a maximum of one-eighth turn of one revolution of the screw. A collar or other means shall be provided to prevent the screw from coming out of the sliding member.

3.5.3 Jaw facings. When specified herein, jaw facings shall be of tool steel or mild steel, milled, or otherwise machined parallel with part of the face that is applied to the castings. Jaw facings of mild steel shall be properly carburized or heat treated to a Rockwell C hardness (HRC) of 52 to 58 (see 4.6.2). Except for type VI, classes 2 and 3, and unless otherwise specified in the contract or order (see 6.2), the main jaw facings shall be either permanently affixed or readily replaceable. Type VI, classes 2 and 3 jaw facings shall be replaceable. All burrs shall be removed and sharp edges broken. The work contact area for the type III vise shall be smooth. All other jaw facings shall be diamond or square serrated.

3.5.4 Vise screw. The vise screw shall be of steel not less than 60,000 pounds per square inch (psi) yield and 80,000 psi tensile strength (see 4.6.3). It shall have American National, square, acme, modified acme, 7/16-14 UNC thread, or buttress threads. The screw threads, except the modified acme thread, or square thread shall be in accordance with FED-STD-H28. One end of the screw shall have a steel or cast iron head of adequate size which shall be an integral part of the screw, welded to the screw, or otherwise securely attached in a permanent manner. The head shall be provided with a hole to receive the screw handle. The screw shall engage the full length of the threaded bearing of the body when the jaw is at the maximum opening indicated in the applicable table.

3.5.5 Nut. The nut, when not an integral part of the body, shall be of malleable iron or heat treated steel, threaded to engage the thread of the screw. Except for types I and VII vises, and when not otherwise specified herein, the nut shall be at least two and one-half times the major diameter of the screw in length. It shall fit properly in the body so that it is in alignment with the screw and there is no binding of the screw in the nut when the jaws of the vise are opened. Any clearance between the nut and the body should not exceed 0.010 inch.

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3.5.6 Screw handle. The screw handle shall be a straight, round steel bar with means provided at each end to prevent complete disengagement and shall slide freely in the hole of the screw head. For types I and VII, the handle shall be of such diameter and length that pressure (within the capacity of the vise) exerted through the screw will not deform the handle. For types III, IV, V, and VI the handle shall be of such diameter and length that it will not be deformed when tested as specified in 4.6.1.

3.5.7 Bench-clamping screw. The bench-clamping screw shall be of a commercial steel screw stock with either a knob wing "T" or sliding bar handle. The clamping screw shall be capable of permitting the vise to be securely tightened to bench tops which vary in thickness from 1/2 to 1-7/8 inches. The upper, or clamping end, of the screw shall have a swivel button to compensate for any irregularities of the bench. The screw threads, except the modified acme thread, or square thread, shall be in accordance with FED-STD-H28.

3.5.8 Swivel base plate. The swivel base plate shall be of the same material as the body, or a steel forging. At least three holes shall be provided for insertion of bench-mounting screws or bolts. The vise, when mounted on the plate, shall swivel through 360 and shall have a positive locking arrangement to secure the vise in any angular position.

3.6 Identification marking. Each vise shall be marked in a plain and permanent manner with the manufacturer's name or identifying symbol, and the state or country of manufacture. The marking shall be engraved, etched, molded, or indented (stamped) directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.7 Type I, bench, clamp base (utility) vise. The type I, bench, clamp base (utility) vise shall consist of a body with back and front jaws, vise screw, screw handle, and bench-clamping screw. The vise shall be provided with guide rods, a single rectangular steel guide bar or vise nut and sliding member. The length of the threaded portion of the nut shall be at least one and one-half times the major diameter of the screw. Guide rods shall be of steel and rigidly attached to the front jaw. The axis of the rods shall be parallel to the axis of the screw. Class 1 and 2 vises shall conform to table I for the size specified (see 6.2).

TABLE I. Type I, class 1 and 2, bench, clamp base (utility), stationary and swivel vise

Jaw width ± 1/8 inch	Jaw opening (minimum)	Throat depth (minimum)	Weight (minimum)	
			Class 1	Class 2
Inches	Inches	Inches	Pounds	Pounds
1-1/2	1-1/2	7/8	2	3-1/4
2	2	1	3-3/8	4-3/4
2-1/2	2-1/4	1-1/4	4	5
3	3	1-1/4	7	10

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3.7.1 Class 1, stationary base. The front of the body shall extend down and form a yoke section with the bottom surface of the body to provide a bench-clamping arrangement. The class 1 vise shall be similar to figure 1.

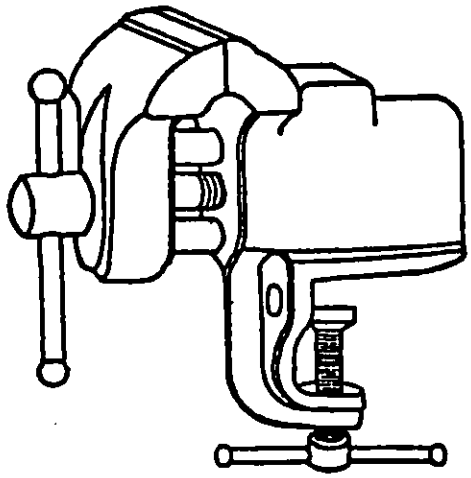


Figure 1. Type I, class 1, bench, clamp base (utility), stationary base vise.

3.7.2 Class 2, swivel base. The class 2 vise shall be similar to figure 2 and shall be designed to swivel on a swivel base plate. The vise, when mounted on the plate, shall swivel through 360 and shall have a positive locking arrangement to secure the vise in any angular position.

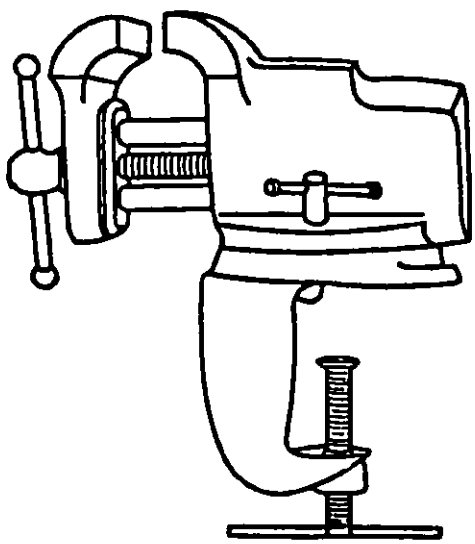


Figure 2. Type I, class 2, bench, clamp base (utility), swivel base vise.

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3.8 Type III, sheet metalworker's vise. The type III, sheet metalworker's vise shall consist of a body, swivel base plate, sliding member, vise screw, nut, and screw handle. The vise shall have high tapering back and front jaws and jaw facings. The vise shall withstand the test loads specified in 4.6.1.

3.8.1 Type III, sheet metalworker's vise. The type III, sheet metalworker's vise shall conform to table II and be similar to figure 8.

TABLE II. Type III, sheet metalworker's vise

Jaw width ± 1/16 inch	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)	
					Cast iron	Steel forging, steel casting, malleable iron; steel cast iron; ductile iron
Inches	Inches	Inch	Inches	Inches	Pounds	Pounds
4-1/2	9	7/8	1-5/8	2-3/16	72	55

NOTE: "Steel cast iron" applies to vises with cast iron bodies and steel sliding bars.

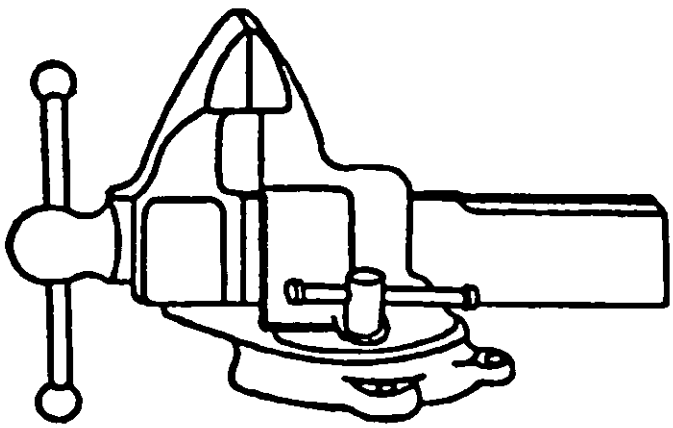


Figure 3. Type III, sheet metalworker's vise.

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3.9 Type IV, bench and pipe (swivel base, stationary jaw) vise. The type IV, bench and pipe (swivel base, stationary jaw) vise shall consist of a body, sliding member, swivel base plate, vise screw, nut, screw handle, pipe jaws, and jaw facings. The vise shall withstand the test load specified in 4.6.1.

3.9.1 Pipe jaws. The pipe jaws shall be of heat treated steel to an HRC of 52 to 58 (see 4.6.2). The jaw shall be located on the front jaw and body of the vise between the main jaw facings and the sliding member. The jaws, when attached to the front jaw and body, shall not project beyond the vertical line of the main jaw facings and shall be easily removed. The face of the jaws shall be V-shaped and have sharp V-serrations to grip the pipe rigidly.

3.9.2 Type IV, bench and pipe (swivel base, stationary jaw) vise. The type IV, bench and pipe (swivel base, stationary jaw) vise shall conform to table III for the size specified (see 6.2) and shall be similar to figure 4.

TABLE III. Type IV, bench and pipe (swivel base, stationary jaw) vise

Jaw width ± 1/16 inch	Nominal pipe size capacity	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)	
						Cast iron	Steel forging, steel casting, malleable iron; steel cast iron; ductile iron
Inches	Inches	Inches	Inches	Inches	Inches	Pounds	Pounds
3-1/2	1/4 to 2-1/2	2	3/4	1-1/4	1-7/8	45	35
4-1/2	1/4 to 3-1/2	5	7/8	1-5/8	2-3/16	75	55
5	1/4 to 4-1/2	6	1	2	2-1/2	115	90
6	1/4 to 6	8	1-1/8	2	2-13/16	175	150

NOTE: "Steel cast iron" applies to vises with cast iron bodies and steel sliding bars.

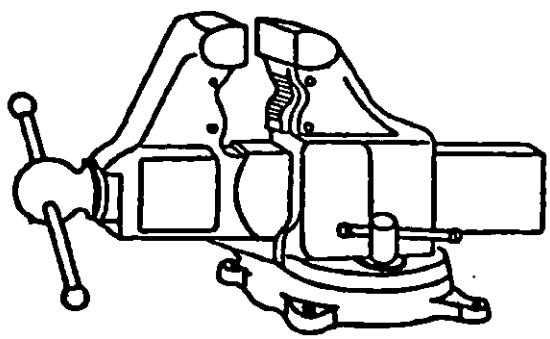


Figure 4. Type IV, bench and pipe (swivel base, stationary jaw) vise.

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3.10 Type V, jeweler's vise.

3.10.1 Class 1, stationary base. The class 1 stationary base shall consist of a body, front and back jaws, jaw facings, sliding member, vise screw, screw handle, and nut.

3.10.1.1 Body. The body shall have a flange or series of lugs around its lower edge which shall be provided with two or more vise mounting screw holes or bolt holes. In all other respects, the body shall conform to 3.6.1.

3.10.2 Class 1, stationary base test load. The class 1, stationary base test load shall withstand the test loads specified in 4.6.1, conform to the sizes specified in table IV, and be similar to figure 5.

TABLE IV. Type V, jeweler's vise

Jaw width ± 1/16 inch	Jaw opening (minimum)	Screw diameter (minimum)	Weight (± 1/4 pound)
Inches	Inches	Inch	Pounds
1-1/4	1	3/8	1
1-1/2	1-1/2	3/8	1-1/2
2	2	3/8	4

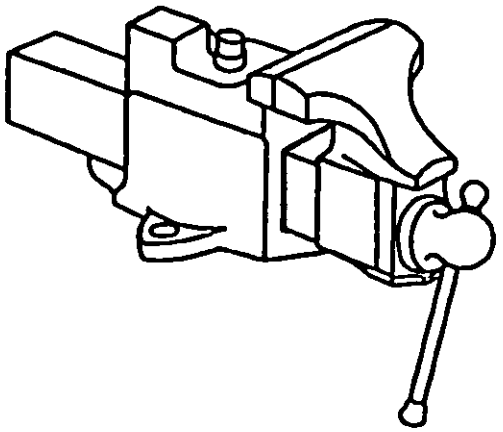


Figure 5. Type V, class 1, jeweler's stationary base vise.

3.10.3 Class 2, swivel base vise. The class 2, swivel base vise shall consist of a body, swivel base plate, jaws, jaw facings, sliding member, vise screw, nut, and screw handle. The vise shall withstand the test loads specified in 4.6.1. The vise shall conform to the sizes specified in table IV and be similar to figure 6.

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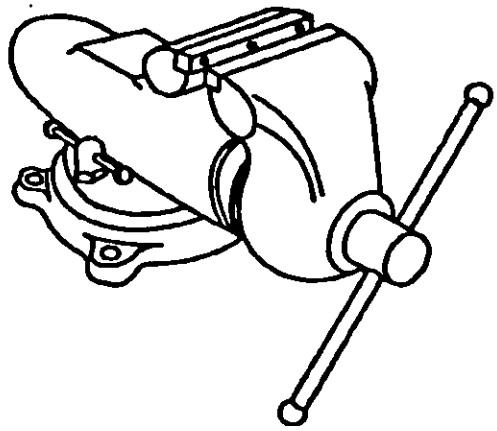


Figure 6. Type V, class 2, jeweler's swivel base vise.

3.11 Type VI, machinist's vise.

3.11.1 Class 2, swivel base, stationary jaw vise. The class 2, swivel base, stationary jaw vise shall consist of a body, front and back jaws, jaw facings, sliding member, vise screw, vise handle, nut, and swivel base. The vise shall withstand the test loads specified in 4.6.1. The vise shall conform to the sizes specified in table V and shall be similar to figure 7.

TABLE V. Type VI, class 2, machinist's, swivel base, stationary jaw vise

Jaw width ± 1/8 inch	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)	
					Cast iron	Steel forging, steel casting, malleable iron; steel cast iron; ductile iron
Inches	Inches	Inches	Inches	Inches	Pounds	Pounds
3	4	5/8	1-1/4	1-9/16	27	25
3-1/2	5	3/4	1-1/2	1-7/8	34	34
4	6	13/16	1-5/8	2-3/16	51	42
4-1/2	6-3/4	7/8	1-5/8	2-3/16	65	55
5	8	1	1-7/8	2-1/2	85	69
6	9	1-1/8	2	2-13/16	150	100
8	12	1-1/4	2-9/16	3-7/16	278	170

NOTE: "Steel cast iron" applies to vises with cast iron bodies and steel sliding members.

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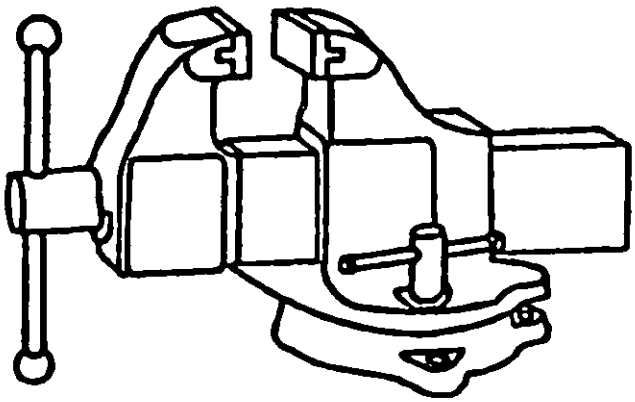


Figure 7. Type VI, class 2, machinist's, swivel base, stationary jaw vise.

3.11.2 Class 3, swivel base, swivel jaw. The class 3, swivel base, swivel jaw vise shall conform to the requirements for class 2 (see 3.11.1) except it shall be provided with a swivel back jaw. The swivel back jaw shall be of the same material as the body and shall be held parallel to the front jaw by means of a steel taper pin. When the pin is removed, the back jaw shall swivel free on a horizontal plane permitting the jaw to align with tapered work. Whether in the stationary or swiveled position, the vise shall clamp the work rigidly. The vise shall withstand the test loads specified in 4.6.1, conform to the sizes specified in table VI, and be similar to figure 8.

TABLE VI. Type VI, class 3, machinist's swivel base, swivel jaw vise

Jaw width ± 1/8 inch	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)	
					Cast iron	Steel forging, steel casting, malleable iron; steel cast iron; ductile iron
Inches	Inches	Inch	Inches	Inches	Pounds	Pounds
3-1/2	4	3/4	1-1/2	1-7/8	36	35
4-1/2	5-1/2	7/8	1-5/8	2-3/16	71	56
6	9	1-1/8	2	2-13/16	164	116

NOTE: "Steel cast iron" applies to vises with cast iron bodies and steel sliding members.

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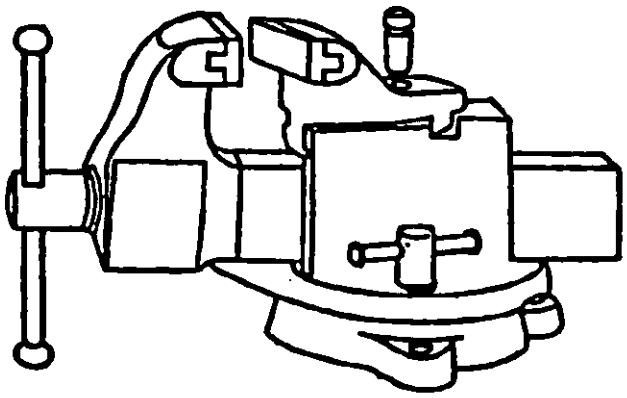


Figure 8. Type VI, class 3, machinist's swivel base, swivel jaw vise.

3.12 Type VII, multiposition bench vise. The type VII, multiposition bench vise shall be so designed and constructed that the work-holding head can be swiveled 360 in a horizontal plane and tilted from a normal level position, downward at least 90, and locked in any desired position within the swiveling and tilting ranges. The vise shall consist essentially of a work-holding head and a base provided with a swiveling, tilting, and locking arrangement. Exclusive of the swiveling, tilting, and locking arrangement, any other joint or fastening between the work-holding head and the base shall be firm and rigid when assembled so as to prevent working loose under normal service conditions.

3.12.1 Work-holding head. The work-holding head shall be in accordance with the type I, class 1, bench vise with a 2-1/2 inch jaw width, as specified in table I, except for the following:

(1) In addition to the metals specified, the body may be of die-cast zinc, die-cast aluminum, or a combination thereof.

(2) In lieu of a clamp base, a shaft bracket, or similar means shall be furnished, for mounting the vise on the swiveling and tilting base.

(3) Jaw facings shall be steel or nylon and serrations are not required.

(4) The nut, when required by the manufacturer's standard design, shall be of malleable iron, steel, brass, or bronze and have a minimum length of 7/8 inch. A nut which is separate from the vise body shall not be required.

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3.12.2 Base. The base shall be provided with a ball and socket arrangement for swiveling the work-holding head, and the socket shall be slotted to provide for tilting the head. The base shall be made of iron, steel, cast aluminum, die-cast aluminum, die-cast zinc, or combinations thereof. The base shall be provided with at least two slots or holes for hold-down screws or bolts. The base shall be provided with
a locking device.

3.12.3 Locking device. The locking device shall lock the work-holding head in any desired position within the swiveling and tilting ranges. The locking device shall operate effectively by one manual operation such as: (1) turning of a screw with hand knob, (2) shifting of a lever, or (3) similar operation.

3.12.4 Design. The type VII multiposition bench vise shall be similar to figure 9.

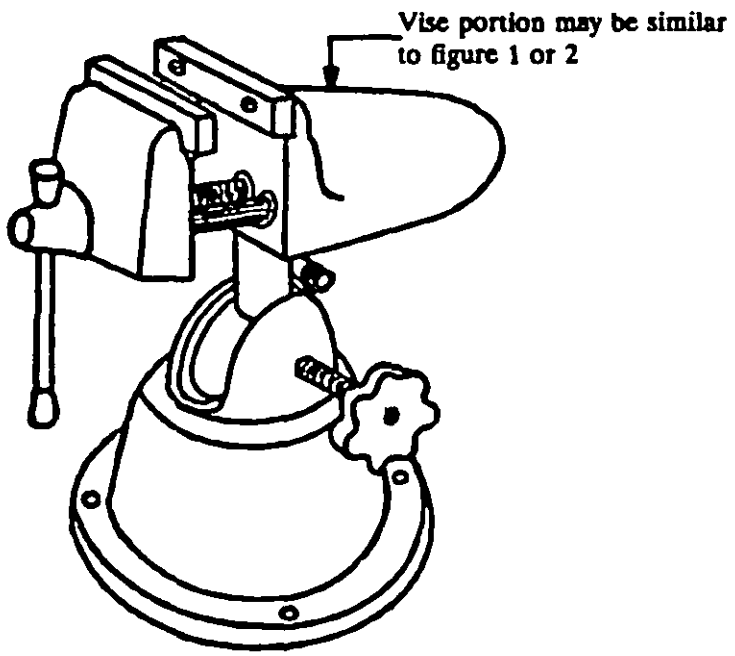


Figure 9. Type VII, multiposition bench vise.

3.13 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

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3.14 Workmanship. Workmanship shall be in accordance with the best grade of commercial practice covering this equipment. All parts shall be free from defects that may affect the serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3 Examination.

4.3.1 Visual examination. The end items shall be examined for the defects listed in table VIII (see 4.7). The lot size shall be expressed in terms of vises. The sample unit shall be one vise. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for total (major and minor combined) defects.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements. The lot size shall be expressed in terms of vises. The sample unit shall be one vise. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4 Testing. The end items shall be tested for the characteristics specified in 4.6.1, 4.6.2, and 4.6.3. The lot size shall be expressed in units of vises. The sample unit shall be one vise. The inspection level shall be S-2 and S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 and 4.0, respectively.

4.5 Back jaw face inspection, type I. Each type I sample vise shall be examined by the inspector to determine conformance with 3.5.1 as follows: The vise shall be set on a surface plate or true plane

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and a protractor shall be used to determine if the back jaw face is within 2 of perpendicularity in either direction.

4.6 Tests.

4.6.1 Loading test. Types III, IV, and VI vises shall be subjected to loading tests to determine conformance with the requirements to table VII. Type V vises shall be subjected to a loading test of 100 inch-pounds for a period of 10 minutes. These tests shall consist of loading the handle as near the end as practicable while the jaws clamp a parallel block of such dimensions that the jaw opening is equal to the nominal width of the jaws specified in tables IV and VII. The handle shall be considered as having taken a permanent set when the maximum deviation of the bent portion is 1/32 inch or more (measurement to be made before and after loading). After the loading test, the vise shall operate smoothly and the component parts shall show no visible distortion.

TABLE VII. Loading test

Jaw width	Test loads, types III and VI	Test loads, type IV
Inches	Inch-pounds	Inch-pounds
3	1,000	
3-1/2	1,200	1,100
4	1,600	
4-1/2	2,000	1,800
5	2,700	2,000
6	3,000	2,200
8	5,000	

4.6.2 Hardness test. A hardness test shall be conducted in accordance with ASTM E18 to determine compliance with 3.5.3 and 3.9.1. Hardness readings for mild steel facings shall be made after removing the outer surfaces. Grinding in preparation for hardness tests shall be performed in a manner to avoid tempering.

4.6.3 Tension test. Each sample vise screw for which the applicable physical requirements of the steel are specified in 3.5.4 shall be tested in accordance with ASTM E 8.

4.7 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VIII. The lot size shall be the

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number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE VIII. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military activities. In addition to any special markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use.

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6.1.1 Type I vise. The type I vises are intended for light bench work and are designed with an integral clamp for attaching to a bench or table.

6.1.2 Type III vise. The type III vises are intended for sheet metalwork and are suitable for bench or table mounting.

6.1.3 Type IV vise. The type IV vises are intended for bench work and are suitable for bench or table mounting. The vises are designed with either integral or detachable pipe-holding jaws.

6.1.4 Type V vises. The type V vises are intended for light bench work and are suitable for bench or table mounting.

6.1.5 Type VI vises. The type VI vises are intended for heavy bench work and are suitable for bench mounting.

6.1.6 Type VII vises. The type VII vises are intended for use in manufacturing and maintenance of mechanical and electronic parts and assemblies. The vises are intended for light bench work which requires multipositioning and quick-locking means.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, and class of vise required (see 1.2.1 and applicable tables).
c. Whether readily replaceable main jaw facings are required (see 3.5.3).
d. Size jaw width or capacity required (see applicable table).
e. Selection of applicable levels of preservation, packaging, and packing required (see 5.1).

6.2.1 Replacement parts. When replacement parts are required, procurement documents should specify the following:

(a) The name or trademark of the manufacturer of the vise for which the part is required.

(b) Proper name or number of the part required.

(c) Quantity of each item required.

6.3 Supersession data. The classification of vises covered in this specification corresponding to those covered in Federal Specification GGG-V-436D are as follows:

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TABLE IX. Supersession data

GGG-V-436D	GGG-V-410
Type I - Bench, clamp base (utility)	Type I - Bench, clamp base (utility)
Class 1 - Stationary base	Class 1 - Stationary base
Class 2 - Swivel base	Class 2 - Swivel base
Type II - Blacksmith's	Deleted
Type III - Sheet metalworker's	Type III - Sheet metalworker's
Type IV - Bench and pipe (swivel base, stationary jaw)	Type IV - Bench and pipe (swivel base, stationary jaw)
Type VI - Jeweler's	Type V - Jeweler's
Type VII - Machinist's	Type VI - Machinist's
Class 1 - Stationary base, stationary jaw	Deleted
Class 2 - Swivel base, stationary jaw	Class 2 - Swivel base, stationary jaw
Class 3 - Swivel base, swivel jaw	Class 3 - Swivel base, swivel jaw
Type XXI - Multiposition bench vise	Type VII - Multiposition bench vise

6.4 Caps. When jaw caps are required, they may be procured under Federal Specification GGG-C-137.

MILITARY INTERESTS:

Army - GL
Navy - SH
Air Force - 69

MILITARY COORDINATING ACTIVITY:

Navy - SH

Preparing Activity:

GSA - FSS

Review Activities:

Army - GL
Navy - SH

User Activities:

Army - MI
Navy - MC, YD
Air Force - 69

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